Several preprocessing steps are usually needed before the image reconstruction can take place. For instance, the logarithm must be taken of each x-ray measurement. This is because x-rays decrease in intensity exponentially as they pass through material. Taking the logarithm provides a signal that is linearly related to the characteristics of the material being measured. Other preprocessing steps are used to compensate for the use of *polychromatic* (more than one energy) x-rays, and *multielement* detectors (as opposed to the single element shown in Fig. 25-14). While these are a key step in the overall technique, they are not related to the reconstruction algorithms and we won't discuss them further.

Figure 25-15 illustrates the relationship between the measured views and the corresponding image. Each sample acquired in a CT system is equal to the sum of the image values along a ray pointing to that sample. For example, view 1 is found by adding all the pixels in each row. Likewise, view 3 is found by adding all the pixels in each column. The other views, such as view 2, sum the pixels along rays that are at an angle.

There are four main approaches to calculating the slice image given the set of its views. These are called **CT reconstruction algorithms**. The first method is totally impractical, but provides a better understanding of the problem. It is based on solving many **simultaneous linear equations**. One equation can be written for each measurement. That is, a particular sample in a particular profile is the sum of a particular group of pixels in the image. To calculate N^2 unknown variables (i.e., the image pixel values), there must be N^2 independent equations, and therefore N^2 measurements. Most CT scanners acquire about 50% more samples than rigidly required by this analysis. For example, to reconstruct a 512×512 image, a system might take 700 views with 600 samples in each view. By making the problem *overdetermined* in this manner, the final image has reduced noise and artifacts. The problem with this first method of CT reconstruction is computation time. Solving several hundred thousand simultaneous linear equations is an daunting task.

The second method of CT reconstruction uses **iterative** techniques to calculate the final image in small steps. There are several variations of this method: the Algebraic Reconstruction Technique (ART), Simultaneous Iterative Reconstruction Technique (SIRT), and Iterative Least Squares Technique (ILST). The difference between these methods is how the successive corrections are made: ray-by-ray, pixel-by-pixel, or simultaneously correcting the entire data set, respectively. As an example of these techniques, we will look at ART.

To start the ART algorithm, all the pixels in the image array are set to some arbitrary value. An iterative procedure is then used to gradually change the image array to correspond to the profiles. An iteration cycle consists of looping through each of the measured data points. For each measured value, the following question is asked: *how can the pixel values in the array be changed to make them consistent with this particular measurement*? In other words, the measured sample is compared with the